

Project Overview

Topic: Neurosymbolic Reasoning

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- 1 Motivation for Neurosymbolic AI
- 2 Problem Statement
- 3 Data
- 4 Related Work
- 5 Challenges & Solutions

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Classical AI: Rule-based/Symbolic AI

Knowledge Base

$\forall x : \text{Human}(x) \rightarrow \text{Wibu}(x)$
(All humans are wibu.)

$\text{Human}(\text{Huy})$
(Huy is human.)

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Deduction Mechanism

$A \rightarrow B$	
A	
$\therefore B$	(Syllogism/Modus ponens)

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Pros

- Explainability
- Data Efficiency
- Abstract Reasoning

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- Brittleness
- Scalability
- Poor Perception

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- Explainability
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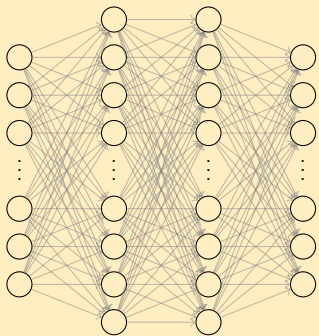
Cons

- Brittleness
- Scalability
- Poor Perception

$\Rightarrow$ **reliable but inflexible.**

Modern AI: Data-driven/Neural Networks

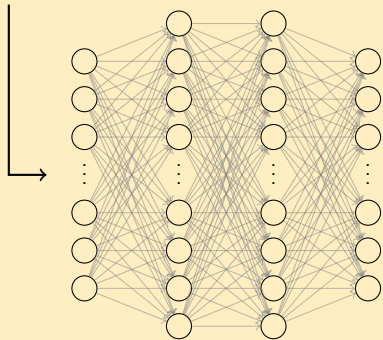
Neural Networks



Modern AI: Data-driven/Neural Networks

Neural Networks

“Is Huy a good boy?”

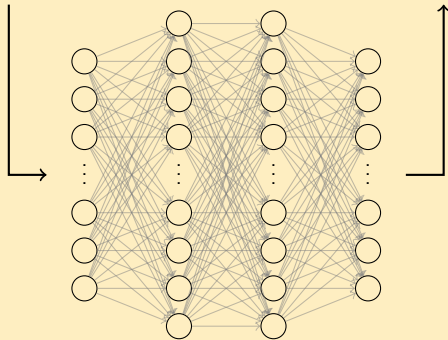


Modern AI: Data-driven/Neural Networks

Neural Networks

“Is Huy a good boy?”

“No, he’s wibu.”

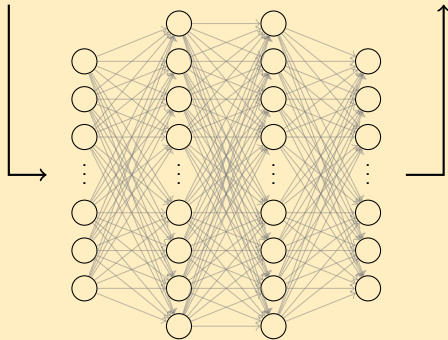


Modern AI: Data-driven/Neural Networks

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Pros

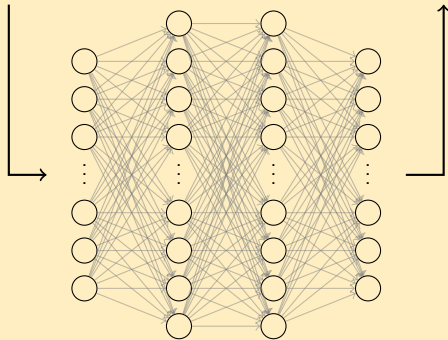
- Generalization & Robustness
- Perception
- Automated Feature Extraction

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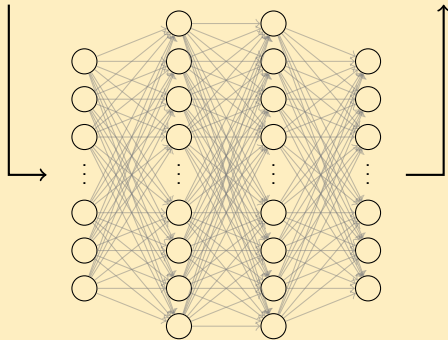
- Lack of Explainability
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- Hallucination

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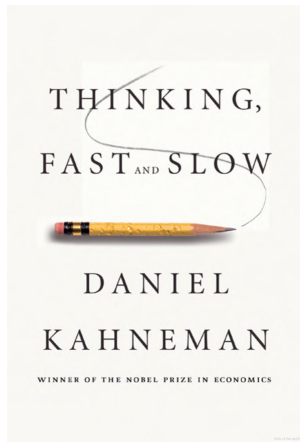
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Cons

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⇒ **flexible but unreliable.**

Neurosymbolic AI: A Hybrid Approach



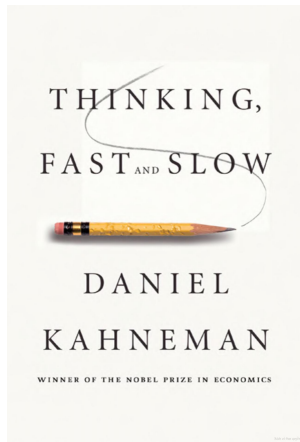
SYSTEM 1 (FAST)

Intuition & Instinct

SYSTEM 2 (SLOW)

Rational thinking

Neurosymbolic AI: A Hybrid Approach



SYSTEM 1 (FAST)

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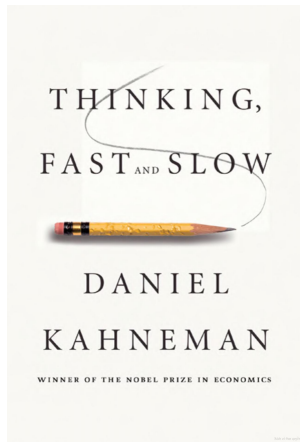
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NEURAL NETWORK

Neurosymbolic AI: A Hybrid Approach



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NEURAL NETWORK

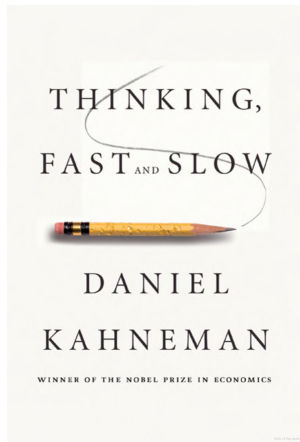
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SYMBOLIC ENGINE

Neurosymbolic AI: A Hybrid Approach



SYSTEM 1 (FAST)

Intuition & Instinct

+

SYSTEM 2 (SLOW)

Rational thinking

=

**HUMAN
INTELLIGENCE**

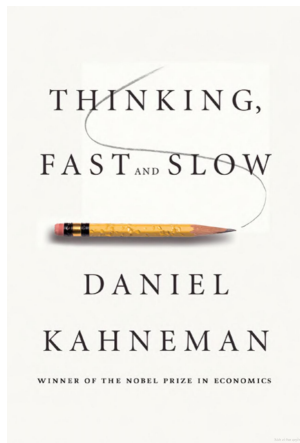
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NEURAL NETWORK

||

SYMBOLIC ENGINE

Neurosymbolic AI: A Hybrid Approach



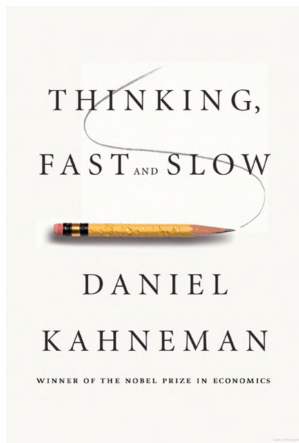
SYSTEM 1 (FAST) + **SYSTEM 2 (SLOW)** = **HUMAN INTELLIGENCE**
Intuition & Instinct Rational thinking

||

||

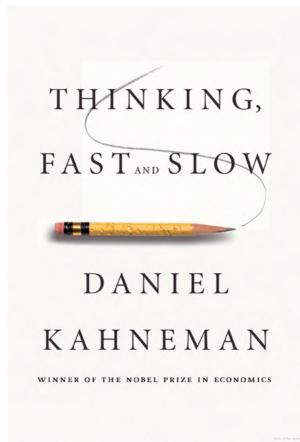
NEURAL NETWORK + **SYMBOLIC ENGINE** = **NEUROSYMBOLIC AI**

Neurosymbolic AI: A Hybrid Approach



NEURAL NETWORK $\oplus$ SYMBOLIC ENGINE = NEUROSYMBOLIC AI

Neurosymbolic AI: A Hybrid Approach



Semantic Parsing/Language Formalization

NEURAL NETWORK $\oplus$ SYMBOLIC ENGINE = NEUROSYMBOLIC AI

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Problem Statement

Language Formalization

The goal is to learn a mapping function $f : \mathcal{X} \rightarrow \mathcal{Y}$ that translates Natural Language sentences (x_{NL}) into machine-verifiable First-Order Logic formulas (y_{FOL}).

Problem Statement

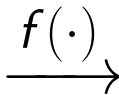
Language Formalization

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Input (x_{NL})

*Natural Language
(Unstructured)*

"Every student takes a course."



Output (y_{FOL})

*First-Order Logic
(Structured)*

$\forall x(\text{Student}(x) \rightarrow \exists y(\text{Course}(y) \wedge \text{Takes}(x, y)))$

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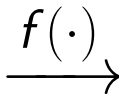
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Core Challenges:

- **The Requirement for Absolute Precision:** Formal languages are syntactically fragile. A formal expression must be 100% correct to be machine-verifiable.
- **Data Scarcity and the Expert-Data Bottleneck:** Creating NL-FOL pairs requires professionals with deep knowledge of both the domain and the formal syntax.

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Data Source

The data used consists of the **training sets** of the datasets listed in the table below.

Source	Volume	Description
FOLIO	1,000 samples	With FOL label , with true/false/uncertain label.
ProofWriter	41,400 samples	Without FOL label , with true/false label, with proof.
RuleTaker	480,000 samples	Without FOL label , with entailment/not entailment label.
ReClor	4,637 samples	Without FOL label , in multiple-choice format.
LogiQA	7,380 samples	Without FOL label , in multiple choice format.

Table: Preliminary statistics of data sources.

Data Source

FOLIO Dataset

FOLIO Dataset

FOLIO is an expert-written, open-domain, logically complex and diverse dataset for natural language reasoning with first-order logic.

Examples:

Premises	Premises-FOL	Conclusion	Conclusion-FOL	Label
All rabbits have fur. Some pets are rabbits.	$\forall x (\text{Rabbit}(x) \rightarrow \text{Have}(x, \text{fur}))$ $\exists x (\text{Pet}(x) \wedge \text{Rabbit}(x))$	Some pets do not have fur.	$\exists x \exists y (\text{Pet}(x) \wedge \text{Pet}(y) \wedge \neg \text{Have}(x, \text{fur}) \wedge \neg \text{Have}(y, \text{fur}))$	Uncertain
All tigers are cats. No cats are dogs. All Bengal tigers are tigers. All huskies are dogs. Fido is either a Bengal tiger or a cat.	$\forall x (\text{Tiger}(x) \rightarrow \text{Cat}(x))$ $\forall x (\text{Cat}(x) \rightarrow \neg \text{Dog}(x))$ $\forall x (\text{BengalTiger}(x) \rightarrow \text{Tiger}(x))$ $\forall x (\text{Husky}(x) \rightarrow \text{Dog}(x))$ $\text{BengalTiger}(\text{fido}) \oplus \text{Cat}(\text{fido})$	Fido is neither a dog nor a husky.	$\neg \text{Dog}(\text{fido}) \wedge \neg \text{Husky}(\text{fido})$	True
...	...	...	...	...

Table: Sample data from the FOLIO dataset.

Data Source

ProofWriter Dataset

ProofWriter Dataset

ProofWriter is ...

Examples:

Data Source

RuleTaker Dataset

RuleTaker Dataset (Rule-based NLI)

RuleTaker is a synthetic dataset for evaluating **multi-hop logical inference** over explicit **facts and rules**.

- **Task:** Given a *context* of English **facts + rules**, predict whether a *question* is **entailed**.
- **Rule form:** conjunctive implication (Horn-like), optionally with negation (e.g., $Kind(x) \wedge Young(x) \rightarrow \neg Smart(x) \Rightarrow$ supports **multi-hop** inference chains.
- **Controlled difficulty:** config (e.g., depth-1, depth-3) indicates reasoning depth.
- **Schema:** context, question, label $\in \{entailment, not\ entailment\}$, config.
- **Use in our pipeline:** large-scale synthetic data (~480k train), mainly for **Phase-2 DPO** (no gold FOL labels).

Examples:

Context (Facts & Rules)	Question	Label	Config
Facts: Bob is kind. Bob is young. Rules: Kind, young things are not smart.	Bob is not smart.	entailment	depth-1
Facts: Fiona is white. Rules: If someone is white then they are not big.	Fiona is not big.	entailment	depth-1
Facts: Fiona is white. Rules: If someone is white then they are not big.	Fiona is big.	not entailment	depth-1

Data Source

ReClor Dataset

ReClor Dataset

ReClor is ...

Examples:

Data Source

LogiQA Dataset

LogiQA Dataset

LogiQA is ...

Examples:

Data Pipeline

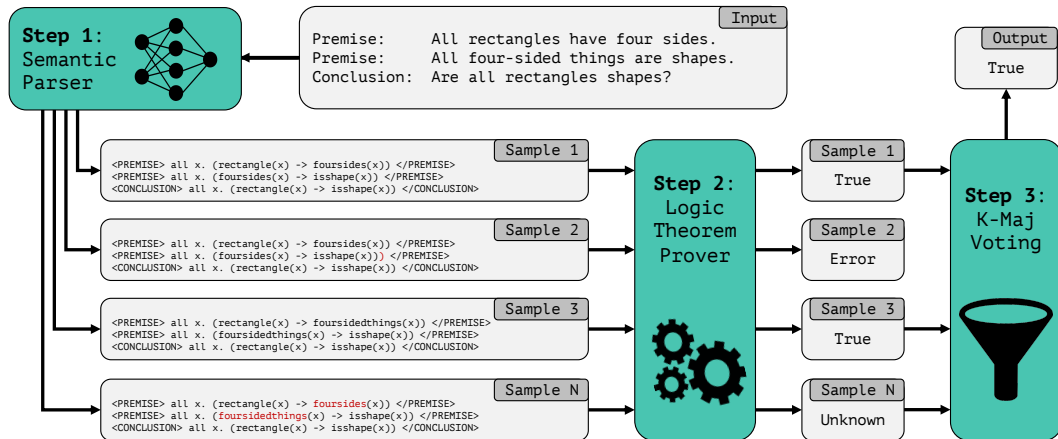


Figure: Mình không hẳn là dùng pipeline này, để tạm đây mai sửa

Data Pipeline

(Viết chi tiết aabcxyz về data pipeline, giai đoạn 1 dùng SFT, giai đoạn 2 dùng DPO)

Supervised Fine-Tuning (SFT):

$$L_{\text{SFT}}(\theta) = - \sum \log \mathbb{P}(y_{\text{FOL}} \mid x_{\text{NL}}).$$

Direct Preference Optimization (DPO):

$$L_{\text{DPO}}(\pi_{\theta}; \pi_{\text{ref}}) = -\mathbb{E}_{(x, y_w, y_l)} \left[\log \sigma \left(\beta \log \frac{\pi_{\theta}(y_w|x)}{\pi_{\text{ref}}(y_w|x)} - \beta \log \frac{\pi_{\theta}(y_l|x)}{\pi_{\text{ref}}(y_l|x)} \right) \right].$$

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Related Work

Liệt kê ngắn gọn đóng góp của các paper liên quan

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Anticipated Challenges & Proposed Solutions

Challenge X

XXX

Solution Y

YYY

Thank you!